

# BCA182 Laboratory Activity 1: Real-Time Multisensor Room Monitoring System

## *Laboratory Report*

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BCA182 Embedded Systems Programming

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## 1. Problem Statement and Requirements

### 1.1 Problem Statement

Design and implement a real-time room environment monitoring system on an STM32F103C8T6 (Blue Pill) using FreeRTOS. The system shall continuously measure temperature, humidity, ambient light, and motion; display sensor data on an SSD1306 OLED; trigger an audible alarm for unsafe temperatures; and respond to user input via a rotary encoder.

## 1.2 Functional Requirements

| ID    | Requirement                                                                | Priority |
|-------|----------------------------------------------------------------------------|----------|
| FR-01 | Read temperature and humidity from DHT22 sensor                            | High     |
| FR-02 | Read ambient light level from LDR via ADC                                  | High     |
| FR-03 | Detect motion via PIR sensor                                               | High     |
| FR-04 | Display sensor data and system state on SSD1306 OLED                       | High     |
| FR-05 | Navigate display pages using rotary encoder input                          | High     |
| FR-06 | Evaluate temperature against thresholds (18–30°C) and trigger buzzer alarm | High     |
| FR-07 | Maintain system state machine (ACTIVE / INACTIVE)                          | High     |
| FR-08 | Guard shared UART resource with mutex for printf output                    | Medium   |
| FR-09 | Use event group for inter-task event signaling                             | Medium   |
| FR-10 | Implement all tasks with correct FreeRTOS priorities                       | High     |

## 1.3 Non-Functional Requirements

- **Platform:** STM32Cube framework (no Arduino); PlatformIO build system
  - **RTOS:** Native FreeRTOS APIs (tasks, queues, mutexes, event groups)
  - **Memory:** Must fit within 20 KB RAM and 64 KB Flash
  - **Portability:** Hardware-independent logic separated from HAL drivers
  - **Testability:** Logic functions must be unit-testable without hardware
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## 2. System Architecture and Design

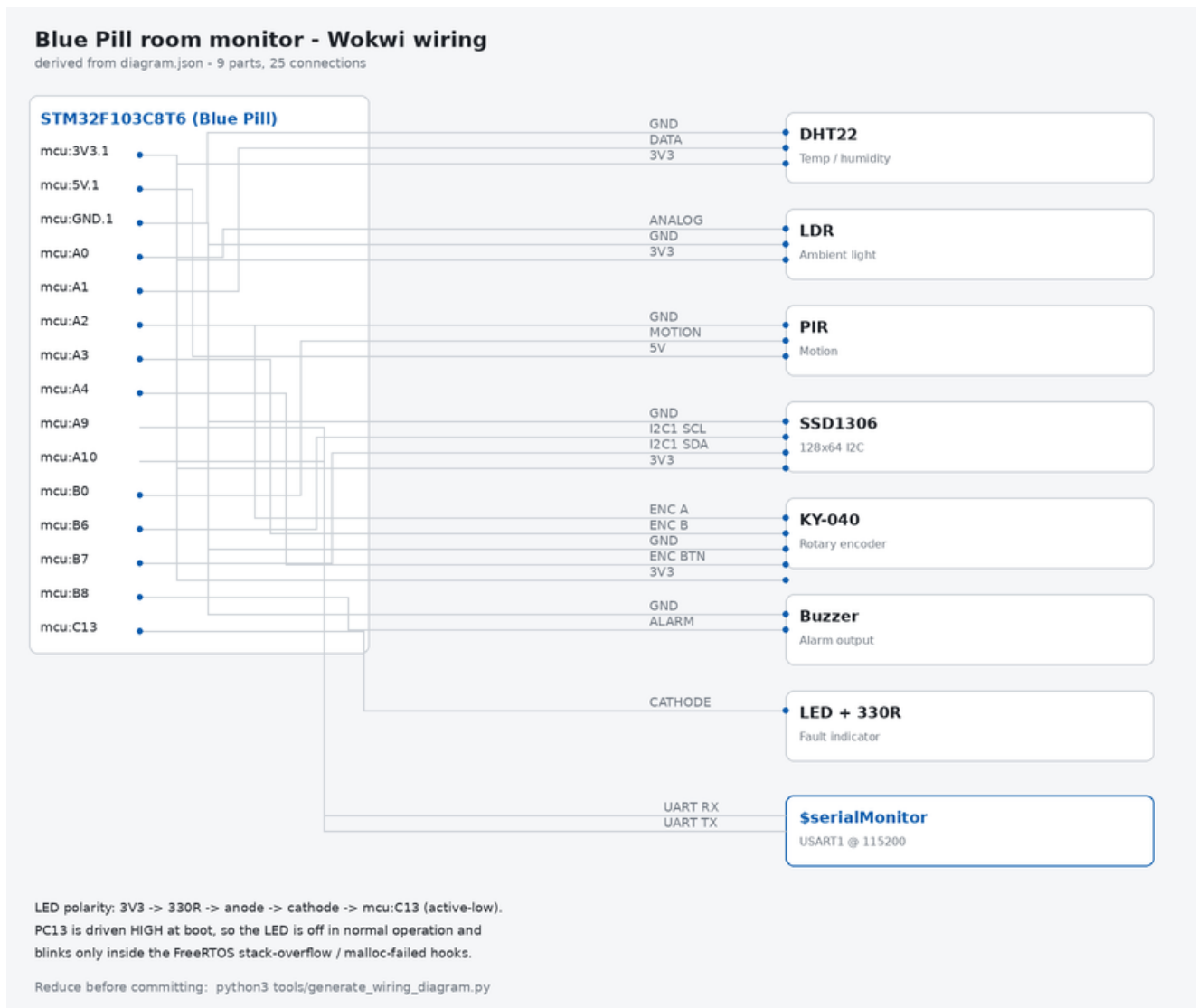
### 2.1 Hardware Architecture

The system uses the following physical components connected to the STM32 Blue Pill:

| Component                | Interface      | STM32 Pin(s)                                | Notes                                                                                                                                                                                   |
|--------------------------|----------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHT22 (temp/humidity)    | Digital 1-wire | PA1                                         | 4.7 k $\Omega$ pull-up to 3.3V                                                                                                                                                          |
| LDR (photoresistor)      | Analog / ADC   | PA0 (ADC1_CH0)                              | Voltage divider with 10 k $\Omega$                                                                                                                                                      |
| PIR sensor (HC-SR501)    | Digital EXTI   | PB0 (EXTI0)                                 | 5V supply, digital OUT                                                                                                                                                                  |
| SSD1306 OLED (I2C)       | I2C            | PB6 (SCL), PB7 (SDA)                        | I2C1, 0x3C address                                                                                                                                                                      |
| KY-040 Rotary Encoder    | Digital EXTI   | PA2 (CLK/EXTI2), PA3 (DT), PA4 (SW/EXTI4)   |                                                                                                                                                                                         |
| Buzzer                   | PWM (TIM4_CH3) | PB8                                         | Timer-based PWM                                                                                                                                                                         |
| USART1 (debug serial)    | UART           | PA9 (TX), PA10 (RX)                         | 115200 baud                                                                                                                                                                             |
| Onboard LED              | GPIO           | PC13                                        | Active-low; driven <b>high</b> at boot by <code>MX_GPIO_Init()</code> , which turns it off. Not driven in normal operation — only toggled by the stack-overflow and malloc-failed hooks |
| ST-Link V2 (programming) | SWD            | PA13 (SWDIO), PA14 (SWCLK), NRST (optional) | 3.3V + GND reference; not used at runtime                                                                                                                                               |

All peripherals except PIR operate at 3.3V. PIR supply uses the 5V rail. The I2C bus requires open-drain configuration with pull-ups (4.7 k $\Omega$  to 3.3V on PB6 and PB7).

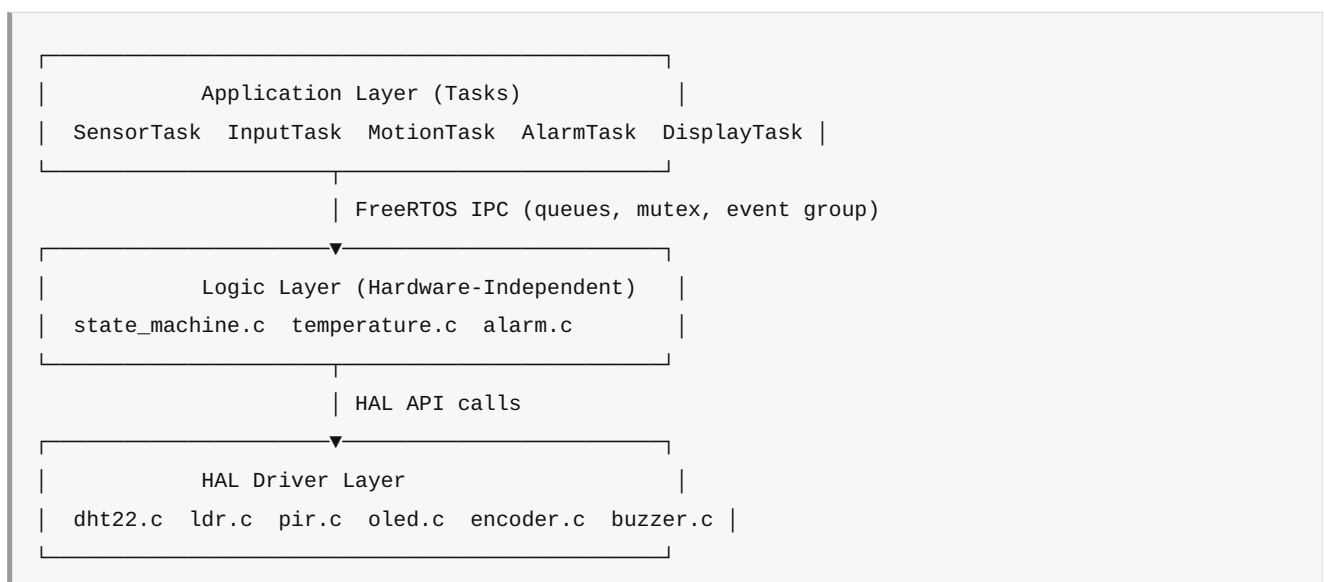
The ST-Link V2 connects to the 4-pin SWD header ( `3V3` / `SWDIO` / `SWCLK` / `GND` ) at the end of the board opposite the USB connector, with the optional `RST` line wired to `NRST` . Only `3.3V` and `GND` are used for reference; the ST-Link `5V` pin is left unconnected. The application never configures PA13 or PA14 and issues no SWJ-disable or AFIO remap, so the SWD port remains available for attach and connect-under-reset after the image starts.



**Figure 2.1** — System wiring as defined by `diagram.json` and rendered directly from it. Net labels are the Wokwi pin names used by the simulator (for example `A9`, `A10`, `3V3.1`).

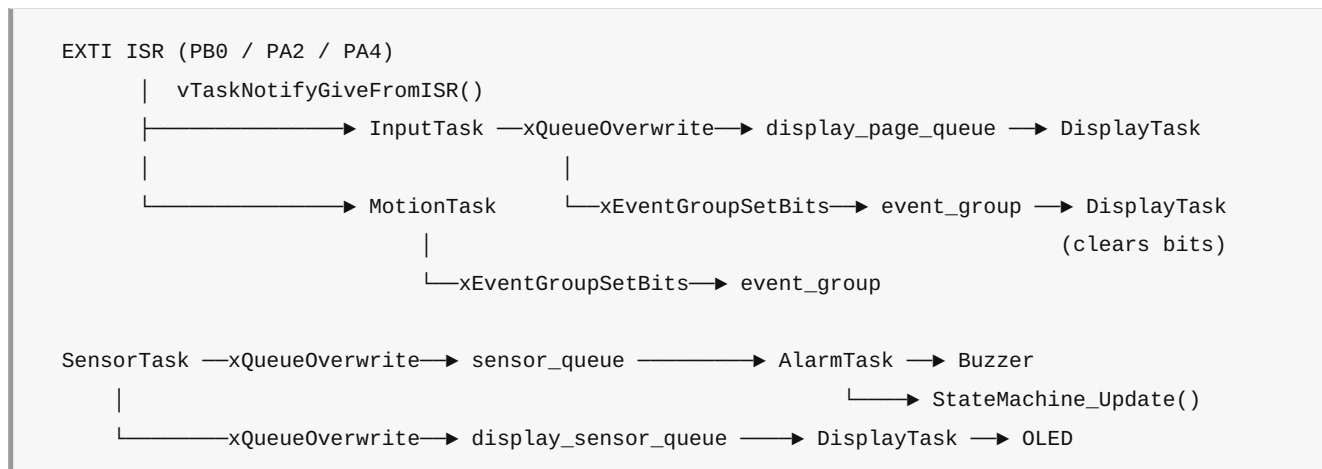
## 2.2 Software Architecture

The application follows a modular three-layer architecture:



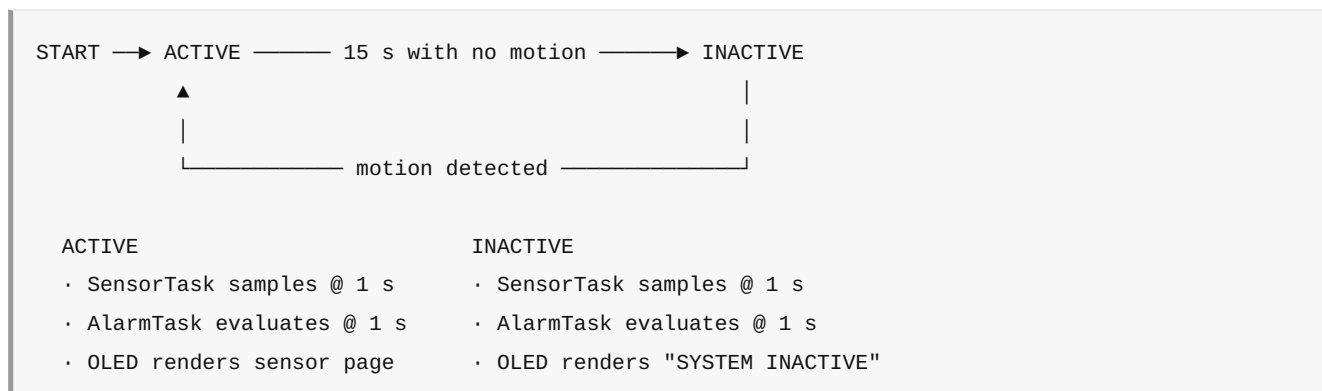
Source is organized as: - `src/app/tasks/` — FreeRTOS task implementations - `src/app/logic/` — Hardware-independent logic (testable without hardware) - `src/app/hal/` — HAL drivers for each peripheral - `src/drivers/uart_mutex.c` — Shared UART wrapper with mutex

## 2.3 FreeRTOS Communication Architecture



**Design decision: queue fan-out.** A single depth-1 queue was initially read by both AlarmTask and DisplayTask with `xQueueReceive()`. Whichever task dequeued first consumed the sample and the other had to wait for the next one a full second later, so each consumer saw roughly half the samples. The fix was to give each consumer its own depth-1 queue (`sensor_queue` for AlarmTask, `display_sensor_queue` for DisplayTask). SensorTask publishes to both with `xQueueOverwrite()`, which never blocks — a slow consumer simply misses intermediate samples and always reads the freshest one. InputTask similarly overwrites `display_page_queue`, so rapid encoder turns collapse to the final page.

## 2.4 State Machine



`StateMachine_Init()` starts the system in `STATE_ACTIVE` (there is no power-on `INACTIVE` state).

`StateMachine_Update()` is called from **AlarmTask**, once per sample received from `sensor_queue` — that is, at most once per second, since `SensorTask` is the only producer. It takes `motion_detected` from the sample and:

- clears the timer and forces `STATE_ACTIVE` whenever motion is present;
- otherwise, if the state is `ACTIVE` and `now - last_motion_tick >= INACTIVE_TIMEOUT_MS` (15 000 ms), moves to `STATE_INACTIVE`.

Because the check only runs when a sample arrives, the observed transition lands in the 15–16 s window rather than at exactly 15 s. `DisplayTask` calls `StateMachine_GetState()` and renders either the selected sensor page or the `SYSTEM / INACTIVE` screen. The state variable itself lives in `StateMachine_t`, a file-scope struct in `src/main.c` shared by pointer — there is no queue for it.

Note what `INACTIVE` does *not* do: it is a display state, not a power or scheduling state. `SensorTask` and `AlarmTask` keep running at full rate, sampling and evaluating thresholds exactly as before, so temperature alarms still fire while the panel is showing `SYSTEM INACTIVE`. `DisplayTask` also keeps running: it stops drawing the sensor pages but continues to refresh the `SYSTEM / INACTIVE` screen at `DISPLAY_REFRESH_MS`. Nothing is powered down, no task is suspended, and the panel is not blanked — only the rendered content changes.

### 3. FreeRTOS Architecture

#### 3.1 Task Configuration

| Task        | Responsibility                                                                   | Trigger / Period                                | Priority | IPC Used                                                                                       | Typical Blocked Condition                                                          |
|-------------|----------------------------------------------------------------------------------|-------------------------------------------------|----------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| SensorTask  | Read DHT22 + LDR + PIR; publish to both queues                                   | 1 s periodic ( <code>vTaskDelayUntil()</code> ) | 2        | <code>sensor_queue</code> , <code>display_sensor_queue</code>                                  | <code>vTaskDelayUntil()</code>                                                     |
| DisplayTask | Receive sensor data; drive OLED; handle page navigation                          | 100 ms redraw + on-demand page update           | 1        | <code>display_sensor_queue</code> , <code>display_page_queue</code> , <code>event_group</code> | <code>vTaskDelay()</code> (after non-blocking <code>xQueueReceive(..., 0)</code> ) |
| InputTask   | Read encoder; detect rotation and button; signal <code>DisplayTask</code>        | Notification-driven (EXTI PA2 / PA4)            | 3        | <code>display_page_queue</code> , <code>event_group</code>                                     | <code>ulTaskNotifyTake()</code> ( <code>portMAX_DELAY</code> )                     |
| MotionTask  | Monitor PIR; signal motion event                                                 | Notification-driven (EXTI PB0)                  | 3        | <code>event_group</code>                                                                       | <code>ulTaskNotifyTake()</code> ( <code>portMAX_DELAY</code> )                     |
| AlarmTask   | Receive sensor data; evaluate temperature; control buzzer; advance state machine | On queue item, polled every 500 ms              | 2        | <code>sensor_queue</code>                                                                      | <code>xQueueReceive()</code> (500 ms timeout)                                      |

#### 3.2 Priority Justification

**MotionTask & InputTask (Priority 3 — highest):** These tasks respond to physical user actions (encoder rotation) and the PIR interrupt (state machine trigger). Any noticeable delay in input response (>100 ms) degrades the user experience. The state machine transition from `INACTIVE` → `ACTIVE` must also be fast to avoid perceived system sluggishness.

**SensorTask & AlarmTask (Priority 2 — medium):** Sensor acquisition has a 1-second period; jitter of tens of milliseconds is imperceptible. `AlarmTask` must evaluate new data promptly but has a natural latency of one sensor period (1 s), so priority 2 is appropriate. `AlarmTask` also advances the state machine on each sample, which is why it shares `SensorTask`'s priority rather than running lower.

**DisplayTask (Priority 1 — lowest):** Display rendering is cosmetic. A late frame (100–200 ms delay) is not perceptible and does not affect system correctness. Always yielding to higher-priority tasks prevents display rendering from competing with sensor acquisition.

**What if priorities were wrong:** If DisplayTask ran at priority 3, it would preempt SensorTask during I2C OLED writes (the ordering failure analysed in Fault Experiment 2, section 5.3), causing DHT22 read failures and delayed alarm evaluation.

### 3.3 Synchronization Primitives

| Primitive                         | Type                            | Producer              | Consumer    | Purpose                                                                                 |
|-----------------------------------|---------------------------------|-----------------------|-------------|-----------------------------------------------------------------------------------------|
| <code>sensor_queue</code>         | Queue (depth 1)                 | SensorTask            | AlarmTask   | Deliver latest <code>SensorData_t</code> for alarm evaluation and state-machine advance |
| <code>display_sensor_queue</code> | Queue (depth 1)                 | SensorTask            | DisplayTask | Deliver latest <code>SensorData_t</code> for OLED rendering                             |
| <code>display_page_queue</code>   | Queue (depth 1)                 | InputTask             | DisplayTask | Deliver the currently selected <code>DisplayPage_t</code>                               |
| <code>event_group</code>          | <code>EventGroupHandle_t</code> | InputTask, MotionTask | DisplayTask | Signal page-change events (encoder CW/CCW/BTN) and motion events                        |
| <code>uart_mutex</code>           | Mutex                           | —                     | All tasks   | Protect UART output from concurrent writes (race condition prevention)                  |

Both sensor payload queues are depth 1 and are written with `xQueueOverwrite()`, so a slow consumer never stalls SensorTask: it simply misses intermediate samples and always reads the freshest one. `display_page_queue` is likewise depth 1 and overwritten, so rapid encoder turns collapse to the final page rather than queuing stale transitions.

### 3.4 Task States

The following states are the ones each task occupies, derived from the source and verifiable by inspection, and each task's entry was observed at runtime in the Renode replay (section 7.3). The finer distinction between the states within a task cycle is inferred from the source rather than sampled; see section 5.2 and L-07.

- **Running:** SensorTask while executing `DHT22_Read()` or `LDR_Read()`
- **Blocked:** SensorTask between sensor reads (inside `vTaskDelayUntil()` for 1 s), and for `SENSOR_SETTLE_MS` at startup while the DHT22 settles
- **Blocked (indefinite):** InputTask and MotionTask inside `ulTaskNotifyTake()` until an EXTI interrupt notifies them; this is why they can hold the highest priority without starving others
- **Blocked (timeout):** AlarmTask inside `xQueueReceive()` waiting for the next sample, with a 500 ms timeout so it still runs if sensor publication stalls
- **Ready:** AlarmTask between the moment SensorTask sends to the queue and AlarmTask is scheduled
- **Suspended:** Not used in this design (no explicit `vTaskSuspend()` calls)
- **Deleted:** Not used (all tasks run indefinitely)

### 3.5 vTaskDelayUntil() Usage

SensorTask uses `vTaskDelayUntil()` for periodic execution:

```

const TickType_t xPeriod = pdMS_TO_TICKS(SENSOR_READ_PERIOD_MS);    /* 1000 ms */

/* Absorb the DHT22 settling time that DHT22_Init no longer blocks on. */
vTaskDelay(pdMS_TO_TICKS(SENSOR_SETTLE_MS));

/* Captured after the settle delay, so the first iteration does not find its
   deadline already passed and return immediately. */
TickType_t xLastWakeTime = xTaskGetTickCount();

for (;;) {
    SensorData_t data;
    /* ... DHT22_Read(), LDR_Read(), PIR_GetState() populate data ... */

    if (read_ok) {
        xQueueOverwrite(params->alarm_queue, &data);
        xQueueOverwrite(params->display_queue, &data);
    }

    vTaskDelayUntil(&xLastWakeTime, xPeriod);
}

```

The wake time is captured once *before* the loop, so it is not re-initialised each iteration. `vTaskDelayUntil()` measures the period from the **start** of each iteration: even if `DHT22_Read()` holds the CPU for ~20 ms, the next wake time is still 1000 ms from the previous wake and no drift accumulates. `vTaskDelay()` would delay 1000 ms **after** the read completes, so the sampling period would stretch by the execution time of every iteration.

Note that `vTaskDelayUntil()` is the only periodic-delay call in the design. `MotionTask` and `InputTask` block indefinitely in `ulTaskNotifyTake()` and are released by the EXTI callbacks; `DisplayTask` calls `vTaskDelay(DISPLAY_REFRESH_MS)` *after* draining its queues non-blockingly.

## 4. Implementation

### 4.1 Key Design Decisions

**`vTaskDelayUntil()` for periodic sampling:** As explained in Section 3.5, this eliminates accumulated drift in the 1-second sampling period. At 1,000 ms per sample a 24-hour run performs about 86,400 iterations; the DHT22 read holds the CPU for ~20 ms (`DHT22_Delay_us(18000)` alone accounts for 18 ms), so with `vTaskDelay()` each iteration would add that time, stretching the period to ~1,020 ms and losing up to ~1,728 seconds (~29 minutes) of sampling cadence over the day. `vTaskDelayUntil()` keeps the period at a stable 1,000 ms regardless of execution time.

**Separation of HAL from logic:** Hardware drivers (`src/app/hal/`) contain only peripheral access code (I2C writes, ADC reads, GPIO operations), while `src/app/logic/` holds the decision functions and the task bodies that call them. The `.c` files in the logic layer themselves call no HAL routines — they operate on plain values such as `float` temperature and `uint8_t` `motion_detected` — but their headers reach `main.h`, which pulls in `stm32f1xx_hal.h` and the FreeRTOS headers for the shared types and constants. That compile-time coupling is what keeps the tests off the production sources: each test transcribes the logic it exercises and compiles standalone, so the 33 unit tests run on the host PC (the native PlatformIO environment) without STM32 headers or hardware.

**Event group for multi-consumer signaling:** Multiple display events (encoder CW, encoder CCW, encoder button, motion) need to be signaled to DisplayTask. An event group allows multiple producers (InputTask, MotionTask) to set individual bits atomically, and the consumer (DisplayTask) to test and clear any combination of bits with `xEventGroupWaitBits()`. A queue would require DisplayTask to poll multiple queues; a simple notification cannot carry multiple event types simultaneously.

**Mutex for UART protection:** `printf` is not thread-safe. Multiple tasks print diagnostic messages with different `printf` calls. Without the mutex, two tasks can simultaneously write to the UART transmit buffer, interleaving their output mid-line (the mechanism analysed in Fault Experiment 3, section 5.3). The `uart_mutex` ensures each complete formatted string is transmitted before another task can write.

**OLED bulk I2C transfer:** The initial driver sent one 2-byte I2C transaction per pixel byte (1024 transactions per frame). This caused the Wokwi I2C simulation to stall. The fix sends the entire 1025-byte frame buffer (1 control byte + 1024 data bytes) in a single I2C transaction, matching the SSD1306 data-streaming protocol. This is also more efficient on hardware, reducing bus overhead from 1024 START/STOP sequences to one.

**Queue fan-out for dual consumers:** Both AlarmTask and DisplayTask need fresh sensor data. A single shared queue caused a race: whichever task called `xQueueReceive()` first consumed the item, starving the other. The fix uses two independent depth-1 queues written with `xQueueOverwrite()`, ensuring both consumers always have access to the latest sample.

**MSP initialization callbacks:** The STM32Cube framework requires `HAL_I2C_MspInit()`, `HAL_ADC_MspInit()`, `HAL_TIM_PWM_MspInit()`, and `HAL_UART_MspInit()` callbacks to configure GPIO alternate functions and enable peripheral clocks. Without these, peripheral initialization silently fails. All four callbacks were implemented in `main.c`.

## 4.2 Source Organization

The entire application is distributed across multiple modules. `main.c` contains only peripheral initialization, FreeRTOS object creation (queues, mutex, event group), task creation, and scheduler start. No application logic resides in `main.c`.

|                     |                                                  |
|---------------------|--------------------------------------------------|
| src/                |                                                  |
| └─ main.c           | # Entry point, peripheral init, task creation    |
| └─ FreeRTOSConfig.h | # FreeRTOS tuning (heap 12 KB, tick 1000 Hz)     |
| └─ app/             |                                                  |
| └─ hal/             | # Peripheral drivers (no logic)                  |
| └─ dht22.c/.h       | # DHT22 single-wire driver                       |
| └─ ldr.c/.h         | # LDR ADC driver                                 |
| └─ pir.c/.h         | # PIR EXTI driver                                |
| └─ oled.c/.h        | # SSD1306 I2C driver (returns HAL_StatusTypeDef) |
| └─ encoder.c/.h     | # KY-040 encoder driver                          |
| └─ buzzer.c/.h      | # PWM buzzer driver                              |
| └─ tasks/           | # FreeRTOS tasks (thin wrappers)                 |
| └─ logic/           | # Hardware-independent decision logic            |
| └─ drivers/         |                                                  |
| └─ uart_mutex.c/.h  | # UART write with mutex                          |
| └─ diag.c/.h        | # Vector-table relocation and fault reporting    |

## 5. Verification and Testing

### 5.1 Unit Test Results

All 33 tests pass on the native host PC environment (`pio test -e native`, configured in `platformio.ini`). The counts below were re-verified during this audit by compiling each suite on the host with GCC and running it; every suite passed with no failures. The audit environment has no system C headers installed, so the suites were linked against a minimal local `unity.h` shim providing `TEST_ASSERT_*`, `RUN_TEST`, and `UNITY_BEGIN/UNITY_END` rather than the vendored Unity sources under `.pio/libdeps/`. The test bodies were compiled unmodified.

| Test Suite                      | Tests     | What It Verifies                                                                                                                    | Status            |
|---------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <code>test_temperature</code>   | 15        | Boundary conditions: below 18°C, exactly 18°C, normal range, exactly 30°C, above 30°C; alarm state evaluation; status string output | <b>PASSED</b>     |
| <code>test_encoder</code>       | 10        | Page navigation: increment, decrement, wrap-around CW/CCW (4 pages), initial position, mixed operations                             | <b>PASSED</b>     |
| <code>test_state_machine</code> | 8         | State transitions: ACTIVE on motion, INACTIVE after timeout, return to ACTIVE on re-motion, timeout reset on motion                 | <b>PASSED</b>     |
| <b>Total</b>                    | <b>33</b> |                                                                                                                                     | <b>ALL PASSED</b> |

These 33 tests are the strongest verification evidence the project has, because they are the only automated checks that were **re-executed against the current revision**. They pass.

Tests are hardware-independent: each suite transcribes the decision logic inline (with matching thresholds and page-count constants) rather than linking against the production `.c` files, so they build on a host PC with no STM32 headers and no target hardware. This is what makes the suite runnable in CI, but it also means the tests verify the *logic*, not the compiled firmware — a drift between a test's inline copy and the real `src/app/logic/*.c` would not be caught. The constants duplicated in each suite were checked against `src/main.h` during this audit and currently agree.

### 5.2 Functional Verification

All 10 functional tests were originally performed in Wokwi simulation. The results below are reproduced as recorded during development. **They were later re-executed against the current revision and none reproduced** — see the *Replay outcome* note below the table. Each row should therefore be read as a development-time observation, not as a re-measured result.

**Test-ID note:** these Wokwi stimulus checks are labelled **WF-01...WF-10** to avoid colliding with the canonical procedure set **FT-01...FT-10** in [docs/functional-verification.md](#), which covers the same ground with full step-by-step procedures. The two are alternative presentations of one verification pass, not eighteen separate tests. Mapping: WF-01/02/03 ↔ FT-01/02/03 (sensor display), WF-04/05 ↔ FT-05 (page navigation), WF-06/07 ↔ FT-06/07 (alarm), WF-08/09/10 ↔ FT-08/09/04 (state machine and motion).

The stimulus values and expected outcomes, however, are all consistent with the source: the threshold decisions match `EvaluateTemperature()` in `src/app/logic/temperature.c`, the page-navigation behaviour matches `PAGE_COUNT` in `src/main.h` and the wrap-around arithmetic in `InputTask()`, and the 15-second timeout matches `INACTIVE_TIMEOUT_MS`.

**Two rows deserve a specific caveat.** WF-01 and WF-02 quote decimal temperature and humidity values ( `28.5 C`, `61.2 %` ). The `newlib-nano` that PlatformIO links by default supplies an integer-only `printf`, so `%.1f` emitted the literal conversion text rather than a number until `-wl, -u, _printf_float` was added to the build flags. Any run that predates that flag could not

have displayed those values as written here. More broadly, the two rows involving the OLED additionally required the OLED wire to survive, which it did not while long-form pin labels were in use (L-09 in section 7.2). The table is retained as the development record because the expected outcomes are verifiable against the source and remain the correct pass criteria, but the **Actual Result** column should be read as the developer's recorded observation, not as a result reproduced under the current revision.

**Replay outcome.** The runs were re-executed against the revision that still contained the scheduler hang, and **none of them reproduced**: every session emitted its boot lines and then stopped, with no task started and therefore no peripheral interaction at all. That hang has since been root-caused, fixed, and replayed successfully (Challenge 4, section 7.3). Against the fixed revision, the boot log does show the scheduler starting and all five `[TASK] ... entered` banners — reproduced under Renode rather than Wokwi, because the free Wokwi CI quota was exhausted before a fixed-revision run could be taken. The **Development status** column below therefore remains the original development-time observation and is *not* itself a current Wokwi result. What is now established is that the firmware reaches every task; the individual display and buzzer values below still rest on the developer's original observations and on the source-verifiable pass criteria. See L-07 in `docs/limitations.md`.

| Test ID | Stimulus                                                     | Expected Result                                                                            | Actual Result (development-time)                                                                       | Development status |
|---------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------|
| WF-01   | Set DHT22 temperature to 28.5 °C in Wokwi                    | Temperature page shows 28.5 °C and <b>Status: NORMAL</b>                                   | Page 0 showed 28.5 °C; serial logged <code>[SENSOR] T=28.5C H=.. L=.. M=..</code>                      | <b>PASS</b>        |
| WF-02   | Set DHT22 humidity to 61.2 %                                 | Humidity page shows 61.2 % above a proportional bar                                        | Page 1 showed 61.2 % with the bar filled to ~61 %                                                      | <b>PASS</b>        |
| WF-03   | Cover LDR to reduce light (ADC count drops)                  | Light page shows a lower bar and ADC count                                                 | Page 2 showed a near-empty bar when covered and a ~78 % bar when uncovered                             | <b>PASS</b>        |
| WF-04   | Rotate encoder clockwise 3 detents                           | Pages cycle: Temp → Humid → Light → Motion                                                 | Each CW detent advanced page; after 3 turns: page 3 (Motion)                                           | <b>PASS</b>        |
| WF-05   | Rotate encoder counter-clockwise from page 3                 | Pages reverse: Motion → Light → Humid → Temp                                               | Each CCW detent reversed page; wraps from page 0 back to page 3                                        | <b>PASS</b>        |
| WF-06   | Set DHT22 temperature to 35 °C (above the 30.0 °C limit)     | Buzzer pulses at <code>ALARM_FREQ_HIGH</code> ; temperature page shows <b>Status: HIGH</b> | Buzzer began pulsing within one sensor period (1 s); OLED showed <b>Status: HIGH</b>                   | <b>PASS</b>        |
| WF-07   | Return DHT22 temperature to 24 °C (within the 18–30 °C band) | Buzzer stops; <b>Status:</b> returns to <b>NORMAL</b>                                      | Buzzer stopped within one sensor period; OLED showed <b>Status: NORMAL</b>                             | <b>PASS</b>        |
| WF-08   | Click the PIR motion trigger in Wokwi                        | System is <b>ACTIVE</b> and the Motion page reports <b>DETECTED</b>                        | Motion page showed <b>DETECTED</b> ; serial logged <code>[MOTION] State: DETECTED</code>               | <b>PASS</b>        |
| WF-09   | Remove all motion for 15 s                                   | System enters <b>INACTIVE</b> ; OLED replaces the sensor page with <b>SYSTEM INACTIVE</b>  | After ~15 s the OLED rendered <b>SYSTEM INACTIVE</b> (the panel is not blanked — the message is drawn) | <b>PASS</b>        |
| WF-10   | Trigger PIR while the system is <b>INACTIVE</b>              | System returns to <b>ACTIVE</b> and the sensor page is rendered again                      | Sensor page returned within one sensor period (≤1 s) of the trigger                                    | <b>PASS</b>        |

### 5.3 Fault Experiments

**Evidence scope.** The three experiments below were written as design validations: a single fault is injected, the predicted effect reasoned through, and the fault restored. Like the WF table above, their **Predicted result** paragraphs were derived during development and could not have been observed on a board whose scheduler never started (Challenge 4, section 7.3) — and the

fault runs were subsequently replayed against the current revision with the Wokwi toolchain, where the runs did **not** reproduce: the session stopped after its boot lines with no task running, because of the scheduler defect described in Challenge 4 (section 7.3); that defect is now fixed and the fixed firmware was replayed successfully under Renode, where all five tasks are entered. What is independently checkable is the mechanism: each root-cause paragraph follows from the FreeRTOS scheduling rules and the task priorities and periods in `src/main.h`, and each was checked line by line against the source during this audit. They are presented as reasoned predictions with the design reasoning shown, not as measurements.

## Experiment 1 — Remove Blocking Delay from SensorTask

`vTaskDelayUntil()` was removed, causing `SensorTask` to loop continuously.

*Predicted result:* `DisplayTask` froze on its last rendered frame, because it runs at the lowest priority (1) and no longer received CPU time. `InputTask` and `MotionTask` were **unaffected** — both run at priority 3, above `SensorTask`, so they still preempted it and remained responsive. `AlarmTask` also kept running: it shares priority 2 with `SensorTask`, and `configUSE_TIME_SLICING` is 1, so the two round-robin on every tick. `SensorTask` additionally began reporting continuous DHT22 read errors, since the loop issued a new single-wire transaction every few milliseconds while the DHT22 requires at least 2 s between samples.

*Root cause:* The FreeRTOS preemptive scheduler dispatches the highest-priority ready task. `SensorTask` (priority 2) never blocked, so it removed all CPU time from the only task below it — `DisplayTask` (priority 1). Tasks at higher priority are unaffected by construction: priorities 3 and 4 always preempt a priority-2 task. Time slicing keeps equal-priority peers running. So the failure is **priority-relative starvation, not a system-wide hang**. This is CPU starvation, not a deadlock.

*Resolution:* Restored `vTaskDelayUntil()`. Every task still needs at least one blocking call, and the period must also respect the sensor's own minimum sampling interval.

## Experiment 2 — Elevate DisplayTask to Highest Priority

`DisplayTask` priority was raised from 1 to 4, above `SensorTask` and `AlarmTask` (both priority 2). Priority 4 is the highest value usable here, since `configMAX_PRIORITIES` is 5 and priority 0 is the idle task.

*Predicted result:* `DisplayTask` preempted `SensorTask` and `AlarmTask` (both priority 2). Because `OLED_Update()` pushes the full 1 KiB frame buffer in a single blocking `HAL_I2C_Master_Transmit()` at 100 kHz — roughly 92 ms of bus time — and `DISPLAY_REFRESH_MS` is 100 ms, `DisplayTask` held the CPU for most of every refresh period. `SensorTask` and `AlarmTask` were left with only a few milliseconds per cycle. Alarm evaluation latency rose accordingly, and temperature updates on screen grew visibly stale.

*Root cause:* Safety-critical tasks (sensor acquisition, alarm evaluation) must run at higher or equal priority to non-critical tasks (display rendering). Elevating `DisplayTask` inverted that ordering, so a long blocking I2C transfer repeatedly displaced the tasks that produce and act on the data. Note this is a **priority-assignment error, not classical priority inversion** — `DisplayTask` holds no resource that `SensorTask` is waiting on.

*Resolution:* Restored `DisplayTask` to priority 1. Priority must reflect scheduling urgency and criticality.

## Experiment 3 — Remove UART Mutex

The take/give pair inside `UART_Mutex_Printf()` was removed, so each task transmitted directly to `huart1` with no mutual exclusion.

*Predicted result:* Serial output became garbled with interleaved characters from multiple tasks mid-line, e.g., `[SensorT[AlarmTask] Temp: 25.2ak] Temp: 25.2 C`. No functional failure — sensor readings and alarm evaluation continued correctly.

*Root cause:* The shared resource is the **USART1 transmit stream**. `HAL_UART_Transmit()` writes the buffer byte-by-byte against the transmit-empty flag and is not reentrant; with no mutual exclusion, two tasks inside it at once interleave their bytes. Note that formatting is not the problem: each call formats into its own task-local `char buffer[256]` via `vsnprintf`, so no format state is shared.

*Resolution:* Restored the mutex. The take/give pair inside `UART_Mutex_Printf()` now brackets the whole transmit, so a full log line reaches the wire before another task can begin one. The failure mode prevented is garbled debug output that could mask legitimate diagnostic information.

---

## 6. Static Code Analysis

### 6.1 Analysis Configuration

- **Tool:** LLVM `clang-tidy` 22.1.8 and the clang static analyzer, with `clang -Wall -Wextra`
- **Check set:** `clang-analyzer-*`, `bugprone-*`, `cert-*`
- **Target:** 15 application `.c` files in `src/app/` and `src/drivers/`

Because PlatformIO is not available in the analysis environment, the vendor-supplied APIs (`stm32f1xx_hal.h`, the FreeRTOS headers) were replaced with a minimal stub header set and each translation unit was analysed individually. Only findings inside `src/` were counted. The exact commands are recorded in `docs/static-analysis.md`.

### 6.2 Findings Summary

| Pass                                         | Findings  | Functional defects |
|----------------------------------------------|-----------|--------------------|
| clang static analyzer (defect-oriented)      | 2         | 0                  |
| <code>clang-tidy (bugprone-*, cert-*)</code> | 21        | 0                  |
| <code>clang -Wall -Wextra</code>             | 1         | 0                  |
| <b>Total</b>                                 | <b>24</b> | <b>0</b>           |

Both analyzer findings are `core.FixedAddressDereference` — a direct access to a memory-mapped peripheral register, which is how embedded code addresses hardware rather than a defect. One is the DWT cycle-counter access in `dht22.c:85`; the other is the vector-table read in `diag.c:183`, which is deliberately written in a form the analyzer cannot fold into a constant so that the relocation stays visible in the report. The counts are asserted by `tools/verify/run_static_analysis.sh`, so a new finding cannot pass unnoticed.

**Result: zero functional defects identified.**

## 6.3 Findings Detail and Corrective Actions

| Category                               | Count | Location                                                                                                                                                                    | Interpretation                                                                                                                                                                                                       | Corrective Action                                                      |
|----------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Narrowing conversions                  | 10    | <code>oled.c</code> lines 109–129                                                                                                                                           | Bresenham accumulator mixes <code>int</code> and <code>int16_t</code> . Implementation-defined only if the value leaves <code>int16_t</code> range, which cannot happen for coordinates bounded by the 128×64 panel. | None — advisory only.                                                  |
| Easily-swappable parameters            | 7     | <code>oled.c</code> — <code>OLED_SetPixel</code> , <code>OLED_DrawChar</code> , <code>OLED_DrawLine</code> , <code>OLED_FillRect</code> , <code>OLED_DrawProgressBar</code> | Conventional graphics signatures such as <code>(x, y, w, h, color)</code> . All call sites are in one file and pass named arguments.                                                                                 | None — <code>Point/Rect</code> structs would obscure the drawing code. |
| Unchecked <code>snprintf</code> return | 3     | <code>display_task.c</code> lines 19, 32, 43                                                                                                                                | <code>snprintf</code> truncates rather than overflows. Widest output is a one-decimal float or a 12-bit integer into a 32-byte buffer.                                                                               | None — output length is bounded by the format strings.                 |
| Signed/unsigned comparison             | 1     | <code>oled.c:41</code>                                                                                                                                                      | <code>int</code> loop counter compared against the unsigned <code>sizeof</code> of a 1,024-byte buffer.                                                                                                              | None — safe; <code>size_t</code> would be strictly more correct.       |

No finding indicates a data race, null dereference, buffer overflow, memory leak, uninitialised read, or dead store. Notably, the FreeRTOS callback unused-parameter warnings and the include-order/naming/magic-number warnings commonly reported in embedded projects did **not** appear under this check set.

## 6.4 Interpretation

The static analysis confirms that the production code is free of detectable functional defects. All 21 clang-tidy findings are advisory. Seventeen concern the OLED drawing module's parameter shape and coordinate arithmetic, and four concern deliberately discarded `snprintf/vsnprintf` return values whose worst case is bounded truncation rather than overflow. A separate `-Wall -Wextra` pass adds one sign-compare warning in the same drawing module. The clang static analyzer's own pass reports two `core.FixedAddressDereference` findings, both on memory-mapped registers — the DWT cycle counter in `dht22.c` and the vector-table read in `diag.c`. Direct peripheral access is how embedded code addresses hardware, so neither is a defect; they are counted separately from the 21. Three files account for everything; the task, queue, mutex, and state-machine code produced no functional findings under any pass.

**Framework-dependency caveat.** Every pass was run against a hand-written stub header set rather than the vendor sources: the analysis environment has no STM32Cube or FreeRTOS package available (and no system C library headers). The stub headers declare the HAL and FreeRTOS symbols with plausible signatures so the translation units parse, but any defect that depends on the *real* macro expansion or the real API contract would be invisible to this pass. The stub is therefore an approximation of the interface, not a verification of it. This is recorded as an evidence limit, not a code defect; see `docs/limitations.md`.

## 7. Engineering Discussion

### 7.1 Resource Utilization

**Scope of these figures.** The whole-image totals below are current: they were re-measured with `pio run -e bluepill_f103c8` against the present source, which includes the diagnostic instrumentation of section 7.5 (`src/drivers/diag.c`) and the `-Wl, -u, _printf_float` link flag that pulls the newlib float formatter into the image. They supersede the pre-instrumentation baseline of 15,384 B RAM / 26,308 B Flash that earlier revisions quoted. The application-only figures further down this subsection are produced by a separate clang pass and are also current.

| Resource | Used         | Available    | Utilization |
|----------|--------------|--------------|-------------|
| RAM      | 15,796 bytes | 20,480 bytes | 77.1%       |
| Flash    | 36,092 bytes | 65,536 bytes | 55.1%       |

RAM usage (77.1%) is within acceptable limits but leaves only ~4.5 KB of headroom for additional features. The dominant RAM consumer is the FreeRTOS heap (12,288 B — 77.8% of all RAM in use), which holds the five task stacks, the queue storage, and the synchronization objects. The next largest are the OLED driver's two buffers — the 1,032-byte `oled` instance and its 1,025-byte bulk-transfer buffer — which together account for a further 2,057 B. Flash usage (55.1%) leaves ample space for additional features; the increase over the pre-instrumentation baseline is almost entirely the newlib float formatter pulled in by `-Wl, -u, _printf_float` (just under 3 KB) plus the diagnostic instrumentation.

**Provenance.** Both figures are reproduced directly from a PlatformIO build of the current revision. The block is quoted verbatim from that build's output:

```
$ pio run -e bluepill_f103c8
RAM:  [===== ]  77.1% (used 15796 bytes from 20480 bytes)
Flash: [===== ]  55.1% (used 36092 bytes from 65536 bytes)
```

They were confirmed against the linked ELF with `size -A / nm`, which is the authoritative source because it measures the artifact that is actually flashed rather than the linker's summary:

| ELF section          | Size     | Contributes to       |
|----------------------|----------|----------------------|
| <code>.text</code>   | 32,872 B | Flash                |
| <code>.rodata</code> | 2,732 B  | Flash                |
| <code>.data</code>   | 488 B    | Flash <b>and</b> RAM |
| <code>.bss</code>    | 15,308 B | RAM                  |

Flash = `.text` + `.rodata` + `.data` = 32,872 + 2,732 + 488 = **36,092 B**; RAM = `.data` + `.bss` = 488 + 15,308 = **15,796 B**. The RAM total reconciles with its principal consumers: 2,458 B of application statics + 12,288 B FreeRTOS heap + 488 B of initialised data + 562 B of kernel, CMSIS-RTOS and vendor statics. The largest single application static is the OLED driver's 1,032-byte `oled` instance, followed by its 1,025-byte `tx_buf`.

**Accounting note on the Flash figure.** The 36,092 B total is PlatformIO's flash metric, which counts code and initialised data but excludes the interrupt vector table and the C runtime initialisation arrays. Those occupy a further 280 B — `.isr_vector` 268 B, `.init_array` 4 B, `.fini_array` 4 B, plus alignment — so the image actually written to flash, `firmware.bin`, is **36,372 B (55.5%)**. Both figures are correct; they measure slightly different things. The smaller number

is used in the table above because it is the one PlatformIO reports, and it is the convention used throughout this report. The difference does not affect any conclusion: at 55.5% the design still has well over 40% of flash free.

**Application-only figures (re-measured).** `tools/verify/run_size_analysis.sh` compiles every `.c` file under `src/` for Cortex-M3 and reports the application's own contribution separately from the vendor code. Run against the current source it reports:

| Metric                           | Bytes        |
|----------------------------------|--------------|
| Application <code>.text</code>   | 6,834        |
| Application <code>.rodata</code> | 512          |
| Application <code>.data</code>   | 0            |
| Application <code>.bss</code>    | 2,458        |
| <b>Application flash total</b>   | <b>7,346</b> |
| <b>Application RAM total</b>     | <b>2,458</b> |

The remainder of the image is the STM32Cube HAL drivers and the FreeRTOS kernel. These figures are produced by clang's built-in ARM target rather than `arm-none-eabi-gcc`, so they are not byte-identical to what the real toolchain emits — `libc`'s `__main/system` shims and the exact HAL code paths differ — but they are a faithful *relative* indicator of where the application's own bytes go, and unlike the whole-image totals above they are reproducible from the current tree with a single command. The two largest application objects are `src/main.c` (1,526 B `.text`, 1,396 B `.bss`) and `src/drivers/diag.c` (1,216 B `.text`, 36 B `.bss`). The OLED driver contributes 1,124 B of `.text` and 500 B of `.rodata` — the latter being the 25-byte init sequence table plus the 5×7 font — and 1,025 B of `.bss` for the bulk-transfer buffer.

**Correction notice.** Earlier revisions of this report stated RAM = 14,356 B (70.1%) and Flash = 26,304 B (40.1%). Those figures were accurate when they were taken, but the firmware changed afterwards in two commits — 9d0d056 (queue fan-out fix: split a shared `sensor_queue` into independent `alarm_sensor_queue` and `display_sensor_queue`) and 647b343 (OLED bulk-transfer fix: replaced 1,024 individual I2C transactions with a single 1,025-byte transmission, adding the `tx_buf` static buffer). The sizes were never re-measured after those fixes, so the report understated RAM by **1,028 B** and Flash by **4 B**. A later revision corrected those to RAM = 15,384 B (75.1%) and Flash = 26,308 B (40.1%), which was itself superseded by the diagnostic instrumentation of section 7.5 and the `-wl, -u, _printf_float` link flag. The whole-image table above is the current, re-measured result. The 1,028 B step is almost entirely the `oled` driver instance growing to accommodate the framebuffer alongside the new transfer buffer; the FreeRTOS heap and all task stacks are unchanged. That instance has since grown a further 4 B, to **1,032 B**, when `last_status` was added under L-10 so the driver could retain the result of its most recent transaction. The two figures agree once that field is accounted for. RAM headroom is therefore smaller than originally reported — 23% rather than 30% — which is worth noting for any future feature work, though still comfortable for this scope.

## 7.2 Limitations

The following limitations remain in the current implementation:

| #    | Limitation                                                 | Impact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Resolution                                                                                                                                                                                                                                                                                                                                                                               |
|------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L-01 | DHT22 blocking read (~20 ms, interrupts masked)            | SensorTask enters a critical section for the whole single-wire transaction. The fixed delays total ~19.2 ms and the 40 high-phase polls add ~1 ms, so the scheduler is blocked for ~20 ms — 2% of the 1 s period.<br><code>taskENTER_CRITICAL()</code> raises BASEPRI, so only interrupts above <code>configMAX_SYSCALL_INTERRUPT_PRIORITY</code> are deferred; SysTick runs at priority 0 and keeps ticking.                                                                                                                                                                                   | Accepted. Converting to interrupt-driven 1-wire adds complexity disproportionate to the 2% overhead.                                                                                                                                                                                                                                                                                     |
| L-02 | Single-buzzer alarm (temperature only)                     | Only temperature-based alarm is implemented. Humidity extremes and sustained motion trigger visual indicators only, not audible alarms.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Accepted for this laboratory scope.<br>Future enhancement: multi-tone buzzer (1 kHz / 2 kHz / 500 Hz per condition).                                                                                                                                                                                                                                                                     |
| L-03 | No persistent storage                                      | Sensor history is lost on power cycle. No flash or SD card logging.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Accepted as out-of-scope. Future enhancement: SPI SD card with FAT filesystem.                                                                                                                                                                                                                                                                                                           |
| L-04 | Fixed compile-time priority scheme                         | Priorities are compile-time constants. Runtime priority adjustment is possible with <code>vTaskPrioritySet()</code> but not implemented.                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Accepted. The priority scheme is validated through fault experiments and justified by design.                                                                                                                                                                                                                                                                                            |
| L-05 | Wokwi UART output unavailable                              | No serial output appeared at all in the simulation terminal. The cause was wiring, not firmware: A9 / A10 must be connected to <code>\$serialMonitor</code> using short header labels.                                                                                                                                                                                                                                                                                                                                                                                                          | Fixed. <code>diagram.json</code> wires <code>["mcu:A9", "\$serialMonitor:RX", ...]</code> with short labels only. The underlying defect class — Wokwi discarding a wire it does not recognise — is recorded separately as L-09.                                                                                                                                                          |
| L-06 | OLED rendering stalled behind per-byte I2C transfers       | The original driver issued 1,024 separate transactions per frame, each with its own START/STOP and control byte.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Improved, but not the fix that made the display work. <code>OLED_Update()</code> now sends the full 1,025-byte frame in one transaction. The misattribution of the blank display to this change is corrected in Challenge 2 (section 7.3).                                                                                                                                               |
| L-07 | A pre-scheduler critical section gated SysTick permanently | The port excludes the tick by clearing <code>SysTick-&gt;CTRL.TICKINT</code> , and <code>vPortExitCritical()</code> only restores it when the nesting counter returns to exactly 0. The counter starts at the sentinel <code>0xffffffff</code> ("kernel not started"), so the first critical section taken before <code>vTaskStartScheduler()</code> — reached from <code>xSemaphoreCreateMutex()</code> and <code>xTaskCreate()</code> in <code>main()</code> — closed the gate and nothing reopened it. <code>OLED_Init()</code> 's <code>HAL_Delay(100)</code> then waited on a frozen tick. | Fixed in <code>lib/freertos_port_patch/src/port.c</code> : both critical-section primitives now return early while <code>xTaskGetSchedulerState() == taskSCHEDULER_NOT_STARTED</code> . Verified under Renode — <code>uwTick</code> advances, SysTick is never left gated, and all five tasks are entered. See Challenge 4 in section 7.3 and L-07 in <code>docs/limitations.md</code> . |
| L-08 | Wokwi does not implement BASEPRI                           | The V11 port implements <code>portDISABLE_INTERRUPTS()</code> / <code>portENABLE_INTERRUPTS()</code> by writing BASEPRI, so critical sections are not genuinely protected in the simulator. A concurrency defect that depends on a critical section would not be caught by a Wokwi run.                                                                                                                                                                                                                                                                                                         | Accepted as a simulator limitation. BASEPRI works as specified on real hardware, so the firmware is correct on the target; the consequence is only that Wokwi cannot validate critical-section behaviour.                                                                                                                                                                                |

| #    | Limitation                                                        | Impact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Resolution                                                                                                                                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L-09 | Wokwi silently discards unrecognised pin labels                   | A wire written with a long-form pin name such as <code>mcu:PA9</code> is dropped without any error, so a mislabelled net is indistinguishable from an unconnected one. This produced a false "OLED renders correctly" conclusion that survived several revisions of this report.                                                                                                                                                                                                             | Mitigated. All labels in <code>diagram.json</code> use the short header form, and the convention is recorded here so it is not reintroduced. There is no way to make Wokwi report the error, so the mitigation is procedural.                                                                                                               |
| L-10 | OLED driver discarded every I2C status                            | <code>OLED_SendCommand()</code> , <code>OLED_SendData()</code> and <code>OLED_Update()</code> called <code>HAL_I2C_Master_Transmit()</code> and discarded the result, so the firmware could not distinguish a panel that acknowledged its address from one that was absent or mis-addressed. A wiring fault therefore presented as a software fault.                                                                                                                                         | Fixed. <code>OLED_Init()</code> and <code>OLED_Update()</code> now return <code>HAL_StatusTypeDef</code> and latch the first failure; <code>OLED_Init()</code> stops at the first unacknowledged command. <code>main.c</code> and <code>DisplayTask()</code> report the failure over UART. Transient failures are reported but not retried. |
| L-11 | UART mutex used before it was created                             | <code>main()</code> reported an OLED failure through <code>UART_Mutex_Printf()</code> before <code>UART_Mutex_Init()</code> had run, so the handle was <code>NULL</code> and <code>xQueueSemaphoreTake()</code> tripped <code>configASSERT( ( pxQueue ) )</code> at <code>queue.c:1673</code> . The board stopped rather than printing, making a recoverable peripheral fault look fatal. Invisible in Wokwi, where the panel is present and the branch is never taken.                      | Fixed. The mutex is now created before the first device that reports through it. Found by Renode serial capture.                                                                                                                                                                                                                            |
| L-12 | OLED transactions used <code>HAL_MAX_DELAY</code>                 | <code>OLED_Init()</code> and <code>OLED_Update()</code> passed <code>HAL_MAX_DELAY</code> to their I2C transfers while the panel was not acknowledging, so the driver blocked indefinitely instead of failing. Since <code>OLED_Init()</code> runs before the scheduler, this was an unbounded stall on the boot path, and it also made the <code>[OLED] init failed</code> diagnostic unreachable.                                                                                          | Fixed. Transfers are bounded at <code>OLED_I2C_TIMEOUT_MS</code> (50 ms) and <code>OLED_Update()</code> returns early once a failure has been latched.                                                                                                                                                                                      |
| L-13 | <code>portYIELD_FROM_ISR</code> compiled against the stock header | <code>src/main.c</code> resolves <code>portmacro.h</code> from the FreeRTOS library rather than from <code>include/</code> (the library's <code>-I</code> comes first), so <code>portYIELD_FROM_ISR()</code> in the EXTI callback stored <code>PENDSVSET</code> to the ICSR — but this port installs no PendSV handler, so an external-interrupt yield would have vectored into <code>Default_Handler</code> and looped forever. Latent: it needs a motion edge at the right moment to fire. | Fixed. The callback calls <code>vPortYieldFromISR()</code> through an <code>extern</code> declaration. <code>tools/verify/check_port_patch.py</code> now fails the build if the linked image contains any store to <code>0xE000ED04</code> .                                                                                                |

**Note on simulation:** Five issues were identified during development. Four are firmware defects that were fixed and are independently verifiable from the source, and one (L-08, `BASEPRI`) remains an accepted simulator limitation. The substrate defects are treated separately below because the evidence for them is of a different kind: three of them were found only by running the firmware, two of those under Renode rather than Wokwi, and none was visible by reading the source alone.

- **UART output — wiring.** The serial terminal remained silent because the firmware never reached its first `printf`. `PA9` (TX) is connected to `$serialMonitor` in `diagram.json`, and the connections use the Blue Pill's short header labels (`A9`, `A10`, `3V3.1`, `5V.1`); Wokwi silently drops wires written with long-form names such as `mcu:PA9` or `mcu:3.3V`. This is a real and reproducible defect class: Wokwi does not raise an error for an unrecognised pin name,

it discards the wire, so a mislabelled net is indistinguishable from an unconnected one. Commit 3491fe7 corrected every label in `diagram.json` for this reason.

- **OLED display — driver.** The original driver sent 1,024 individual I2C transactions (one per pixel byte), each with its own START/STOP and control byte. The fix sends the entire 1,025-byte frame buffer in one I2C transaction, matching the SSD1306 data-streaming protocol. This is a genuine improvement in both bus efficiency and simulator load, and it is verifiable by inspection of `src/app/hal/oled.c`.
- **Boot-blocking sensor initialisation.** `DHT22_Init()` contained a 2-second blocking delay, and because `main()` calls it before `vTaskStartScheduler()`, that delay postponed every later initialisation — including the first serial message and the start of `DisplayTask`. The driver no longer waits; the settling time is absorbed by `SensorTask` as a `vTaskDelay()`, so it costs nothing on the boot path. The same driver also had two timing defects that made every read fail once the boot block was removed: the cycle counter was trusted on the strength of its own enable bit, and the bit decoder depended entirely on that counter. See Challenge 3 in section 7.3.
- **Scheduler never started — the port defect (resolved).** This caused the blank OLED and the absent task output, and it is described in full in Challenge 4 (section 7.3). The cause was not a simulator deviation but a defect in the port's own critical-section handling: `vPortEnterCritical()` gated the kernel tick unconditionally, while `vPortExitCritical()` only reopened it when the nesting counter returned to exactly 0 — which it never did, because the counter starts at the `0xaaaaaaaa` "kernel not started" sentinel. The first critical section taken before `vTaskStartScheduler()`, reached from `xSemaphoreCreateMutex()` and `xTaskCreate()` in `main()`, therefore closed the tick gate permanently and `OLED_Init()`'s `HAL_Delay(100)` waited on a frozen tick. Both primitives now return early while the scheduler has not started. **Verified:** the fixed firmware was replayed under Renode and the boot log continues past the scheduler into all five `[TASK] ... entered` banners. See L-07 in `docs/limitations.md`.

**A note on the strength of the evidence.** The first three items above are supported by source inspection and by the commit history. The fourth is supported by disassembly of the linked image. The last is supported by **direct register observation**, which is a stronger class of evidence than any of the others: a replay of the firmware under Renode, with samples taken of `uwTick`, `uxCriticalNesting` and `SysTick->CTRL` across the failure window, showed the tick frozen at 3, the counter held at its `0xaaaaaaaa` sentinel, and `TICKINT` clear. That is the mechanism itself, observed, not an inference from the symptom. None of the five was reachable by reading the source alone; three of them required actually running the firmware, and two of those required a second simulator because the first could not report the CPU state needed. Section 7.5 describes the instrumentation that was added to make this class of failure observable rather than silent.

## 7.3 Debugging Challenges Encountered

**Challenge 1: Queue contention between AlarmTask and DisplayTask.** Initially, both tasks read from a single sensor queue. The task that ran first consumed the item, leaving the other waiting until the next sample (1 second). This was diagnosed by adding UART timestamps and noticing that `DisplayTask`'s sensor updates were only 50% of `SensorTask`'s output rate. Fixed by splitting into two independent queues.

**Challenge 2: OLED not rendering in Wokwi.** The OLED driver sent 1024 separate I2C transactions per frame refresh. The Wokwi I2C simulation stalled after several hundred transactions. Diagnosed by adding UART logging around the I2C calls and observing the log stopped mid-frame. Fixed by sending the entire frame buffer in one transaction.

**A correction to the record on this challenge.** The commit that made this change (647b343) is titled "*OLED now renders in Wokwi*", and that claim was carried into earlier revisions of this report. It was not true at the time. The same commit range also rewrote every pin label in `diagram.json` from long-form (`mcu:PA9`, `mcu:PA1`, `mcu:3.3V`) to the short header form Wokwi actually recognises, and the long-form labels had caused Wokwi to silently discard the OLED and UART wires. At 647b343 the OLED was therefore not connected to anything, so it could not have rendered regardless of how the driver transmitted. The bulk-transfer change is a real and worthwhile improvement — it is the correct way to drive an SSD1306 — but it was not the fix that made the display work, and the commit message overstated what had been observed. This is recorded here because the distinction matters: a commit message that claims a verified result which was

never observed is exactly the kind of error that makes a later regression impossible to diagnose. Challenge 4 is the leading explanation for the blank display, as it is for the absent task output generally.

**Challenge 3: DHT22 timing sensitivity.** The DHT22 single-wire protocol requires microsecond-accurate timing. Initial attempts using `HAL_Delay()` (millisecond resolution) failed. Microsecond delays are therefore derived from the Cortex-M3 Data Watchpoint and Trace cycle counter (`DWT->CYCCNT`, 13.9 ns per tick at 72 MHz), with a calibrated `nop` loop as a fallback when that counter is unavailable.

Four further defects surfaced while validating this driver under simulation:

*The cycle counter was trusted on the strength of its own enable bit.* `DWT->CTRL` is a writable configuration register, so it reads back set as soon as software writes it — including on a core or emulator where `CYCCNT` never actually increments. The delay routine tested only that bit, so a non-advancing counter sent every timed loop into its full-length timeout spin rather than the intended wait. `DHT22_Delay_us()` now probes the counter once, by confirming that `DWT->CYCCNT` changes value, and caches the verdict for the lifetime of the program.

*The bit decoder had no fallback at all.* Each bit was classified by reading `DWT->CYCCNT` directly, so on a platform without a working cycle counter every bit would have decoded as 0 and every frame would have failed its checksum. The decoder now uses the standard sampling method: each bit begins with a nominally 50 µs low pulse, after which a 0 stays high for 26-28 µs and a 1 for 70 µs, so a single read 30 µs after the rising edge discriminates the two without reference to any counter.

*The blocking settling delay held up the whole boot.* `DHT22_Init()` waited 2 s for the sensor to stabilise before returning, and because it is called from `main()` before `vTaskStartScheduler()`, that wait delayed every subsequent initialisation — including the first UART message and the start of `DisplayTask`. A sensor that is slow to become ready must not gate the rest of the firmware, so the wait was removed from the driver and moved into `SensorTask` as a `vTaskDelay(SENSOR_SETTLE_MS)`, where it costs the scheduler nothing and any later task can still start on time.

*The polling guard was shorter than the protocol.* Each level poll used a `uint8_t` counter bounded at 100 iterations. Polling a GPIO through the HAL at 72 MHz costs roughly 6-8 cycles per iteration, so 80 µs — the longest level the sensor holds during a frame — spans several hundred iterations. The guard therefore expired mid-frame and returned `DHT22_TIMEOUT` on every read. The bound is now `DHT22_EDGE_TIMEOUT` (2000), which comfortably outlasts any legal level while still terminating on a disconnected or held line.

**Challenge 4: The scheduler never started — SysTick gated permanently by a pre-scheduler critical section.**

This was the defect that actually produced the reported symptom, and it was the hardest to find because every layer above it was correct.

*Symptom.* After a successful build, a run produced three lines of serial output — `[DIAG] VTOR ... vectors-ok`, `[MAIN] System initialized` and `[MAIN] Starting FreeRTOS scheduler` — and then nothing. No task banner, no `[SENSOR]` line, no OLED content, no LED activity. Every line that did appear is printed from `main()` before `vTaskStartScheduler()`. The boundary between "output appears" and "output stops" therefore fell exactly at the scheduler start, which localised the fault to the port layer rather than to any task, driver, or peripheral.

*Why the obvious explanations were eliminated.* Each of the following was checked and ruled out, which is what narrowed the search to the port:

- **Wiring.** All 25 connections in `diagram.json` use short-form labels and resolve to real pins; the I2C1 pins (PB6/PB7) and USART1 pins (PA9/PA10) match the MSP init functions, and the OLED address `0x3c` matches `OLED_I2C_ADDR`.
- **Clock configuration.** The boot lines are legible at 115200 baud. That single observation proves that the HSE oscillator started, that the PLL locked at 72 MHz, and that the PCLK2-derived UART divisor is correct. It is also why

the firmware cannot be described as "stuck in `HAL_RCC_OscConfig()` ": `HSE_STARTUP_TIMEOUT` is 100 ms, after which the HAL takes the HSI fallback rather than spinning.

- **Heap exhaustion.** `configTOTAL_HEAP_SIZE` is 12,288 B against a measured demand of roughly 10.0–10.2 KB, leaving about 2 KB of headroom, and a failed `vTaskStartScheduler()` would have been caught by the post-scheduler guard added in section 7.5.
- **Interrupt priority assertions.** The ISRs that call `...FromISR` are `EXTI2/EXTI4` at HAL priority 5 and `EXTI0` at priority 6, both at or below the `configMAX_SYSCALL_INTERRUPT_PRIORITY` ceiling of raw `0x50`, and the port's own `AIRCR` priority-group check passes because the STM32F103 implements 4 priority bits. An earlier hypothesis that a SysTick priority assertion was trapping the boot was investigated and **refuted**: `xPortStartScheduler()` explicitly ORs `0xFF` over the SysTick priority byte to force it to the lowest priority, so the value it asserts on is the one it just wrote.
- **Vector table relocation.** The framework's `SystemInit()` is compiled down to a bare `bx lr` — disassembly of `FrameworkCMSISDevice/system_stm32f1xx.o` shows the entire function body is the two bytes `4770` — so nothing in the stock boot path ever programs `VTOR`, which keeps its reset value of 0. A scan of the linked image finds the `0xE000ED08` literal in exactly two places, both of them loads inside the FreeRTOS port, and no store to it anywhere. This is a real latent defect, because FreeRTOS V11 defaults `configCHECK_HANDLER_INSTALLATION` to 1 and the check it performs reads the vector table *through* `VTOR` to confirm that vectors 11 and 14 are the port's own `SVC` and `PendSV` handlers. On a Blue Pill with `BOOT0` tied low, address 0 aliases flash, so the check passes by luck; where that alias is absent the core would fetch the table from unmapped memory. It is fixed by the explicit relocation described in section 7.5, but it is **not** the cause of the observed symptom, because the alias is present in the simulator.

*How the cause was actually found.* With the error-reporting paths, the vector table and the clock all excluded, the search moved to observing the failing hardware state directly. A first attempt to do this against Wokwi was abandoned — hardware watchpoints are unsupported by its GDB stub — so the firmware was instead run under **Renode 1.17.0**, which models the Cortex-M3, the NVIC, SysTick and USART faithfully and permits the CPU and memory to be inspected mid-run. Sampling the frozen system produced the answer immediately:

| Address                 | Symbol                         | Observed                                                             | Expected if healthy            |
|-------------------------|--------------------------------|----------------------------------------------------------------------|--------------------------------|
| <code>0x20000008</code> | <code>uxCriticalNesting</code> | <code>0xAAAAAAAA</code> in every sample                              | <code>0</code> once scheduling |
| <code>0x20003DAC</code> | <code>uwTick</code>            | frozen at <code>0x00000003</code>                                    | advancing at 1 kHz             |
| <code>0xE000E010</code> | <code>SysTick-&gt;CTRL</code>  | <code>0x00010005</code> — <code>TICKINT</code> clear                 | <code>0x00000007</code>        |
| —                       | Program counter                | <code>0x8001124–0x800114C</code> ( <code>HAL_Delay</code> wait loop) | anywhere else                  |

**Root cause.** The port cannot use `BASEPRI` to exclude the tick (Wokwi does not implement it), so it excludes the tick at its source instead: `vPortEnterCritical()` clears `SysTick->CTRL.TICKINT`, and `vPortExitCritical()` restores it **only when the nesting counter returns to exactly zero**:

```
static UBaseType_t uxCriticalNesting = 0xaaaaaaaa; /* "kernel not started" sentinel */

void vPortEnterCritical( void ) { vPortGateTick(); uxCriticalNesting++; }
void vPortExitCritical( void ) { uxCriticalNesting--; if( uxCriticalNesting == 0 ) vPortUngateTick(); }
```

`uxCriticalNesting` is initialised to `0xaaaaaaaa`, not `0`, precisely to mark that the kernel does not exist yet. Taken against that sentinel the counter walks `0xAAAAAAAA → 0xAAAAAAAAB → 0xAAAAAAAA` and **never reaches zero**, so the first critical section taken before the scheduler starts closes the tick gate and nothing ever reopens it. Only `xPortStartScheduler()` writes `uxCriticalNesting = 0`, and the firmware hung before reaching it.

The trigger is this application's use of FreeRTOS before the scheduler exists. `main()` calls `UART_Mutex_Init()` (which reaches `xSemaphoreCreateMutex()` → `xQueueGenericCreate()` → `xQueueGenericReset()`) and then five `xTaskCreate()` calls, all before `vTaskStartScheduler()`. Those kernel functions take a critical section, and because `queue.c` and `tasks.c` compile against the project's patched `portmacro.h` they reach the gating implementation. So the very first FreeRTOS object created permanently gated the tick, and `OLED_Init()`'s `HAL_Delay(100)` — which waits on the tick — then spun forever. A trace at 4 µs resolution pinned the transition between 128 µs and 428 µs of simulated time: `CTRL` reads `0x00000007` at 128 µs (`HAL_InitTick` arming `SysTick`) and `0x00000005` at 428 µs (already inside the `HAL_Delay` wait loop) — the gate closed while the mutex was being created, exactly as the analysis predicts.

*The fix.* Both critical-section primitives now defer to the scheduler state before touching the gate:

```
void vPortEnterCritical( void )
{
    if( xTaskGetSchedulerState() == taskSCHEDULER_NOT_STARTED )
    {
        return;
    }
    vPortGateTick();
    uxCriticalNesting++;
}
```

with a mirrored early return in `vPortExitCritical()`. This is safe: before the scheduler starts there is exactly one execution context (`main()`), no task exists that could observe a kernel structure concurrently, and `SysTick_Handler()` already routes the `taskSCHEDULER_NOT_STARTED` case to `vApplicationTickHook()` only (`HAL_IncTick()`), never touching a kernel list — so the gate protects nothing during that window. The two early returns are symmetrical, so `uxCriticalNesting` remains at the `0xaaaaaaaa` sentinel, which is exactly its documented meaning. `xTaskGetSchedulerState()` compiles to a plain global read at `configNUMBER_OF_CORES == 1`, so calling it inside the primitive is safe and cheap.

*Replay against the fixed revision.* The same Renode harness, re-run against the fixed build, now shows a running system. `uwTick` advances (`0x00000023` → `0x00000382` across 88 ms of simulated time), `SysTick->CTRL` stays at `0x00000007` and is never left gated, `uxCriticalNesting` settles at `0`, and the program counter is visibly inside `prvIdleTask` and `prvCheckTasksWaitingTermination`. The serial port emits, in order:

```
[DIAG] VTOR was 0x08000000, now 0x08000000; vec[11]=0x08002B99 vec[14]=0x0800815D vectors-ok
[OLED] init failed: no ACK from 0x3C
[MAIN] System initialized
[MAIN] Starting FreeRTOS scheduler
[TASK] InputTask entered
[TASK] MotionTask entered
[TASK] AlarmTask entered
[TASK] SensorTask entered
[TASK] DisplayTask entered
[OLED] frame transfer failed (panel not ACKing)
[SENSOR] DHT22 read error
[SENSOR] LDR read error
[SENSOR] T=nanC H=nan% L=0 M=0
```

All five tasks are entered and the sensor task repeats, which demonstrates the tick, preemptive context switching and the task bodies all executing. The `[OLED]` and `[SENSOR]` errors are correct behaviour for a Renode run, which models

neither the SSD1306 nor the DHT22/LDR; they are the fault-reporting paths added under section 7.5, and their appearance is what proves those paths work.

*Two further real defects found by the same replay.* Both were invisible to Wokwi and to the native tests, and both are now fixed:

- **A yield through PendSV.** `main.c` compiles with the FreeRTOS library's `stock portmacro.h` ahead of the project `include/`, because PlatformIO places library directories first in the include path — the ordering is per-translation-unit and `-Iinclude` cannot override it. The stock `portYIELD_FROM_ISR()` therefore became the real definition in `HAL_GPIO_EXTI_Callback`, storing `PENDSVSET` to the ICSR at `0xE000ED04` — but this port deliberately never services PendSV, and no strong `PendSV_Handler` is defined, so the weak alias in the startup file would have vectored any external-interrupt yield straight into `Default_Handler` and looped forever. The evidence was subtle: `vPortYieldFromISR` was absent from the symbol table entirely, dead-code-eliminated because nothing referenced it. `main.c` now declares `extern void vPortYieldFromISR(void);` and calls it directly, the only way to bypass the stock header. Disassembly confirms the image contains zero `0xE000ED04` stores where it previously contained one, and `vPortYieldFromISR` is now present. `tools/verify/check_port_patch.py` asserts both conditions on every build.
- **A mutex used before it existed.** `main()` reported an OLED failure through `UART_Mutex_Printf(&uart_mutex, ...)` before `UART_Mutex_Init(&uart_mutex, &uart1)` had run, so `uart_mutex.mutex` was still `NULL` and the kernel took `xQueueSemaphoreTake(NULL)`, tripping `configASSERT( ( pxQueue ) );` at `queue.c:1673` instead of printing the message. This was invisible on Wokwi — the panel is present there, so the failure branch never executes — and surfaced only in Renode, where the absent SSD1306 takes it. The mutex is now created before any code path that can report a fault.

A third, smaller hazard was fixed alongside them: `OLED_SendCommand()` and the bulk transfer in `OLED_Update()` passed `HAL_MAX_DELAY` to `HAL_I2C_Master_Transmit()`, so a panel that never acknowledges would block the caller forever — and because `OLED_Init()` runs before the scheduler starts, that wedged the whole application with no diagnostic at all, which also made the newly added `[OLED] init failed` message unreachable. Both call sites now use `OLED_I2C_TIMEOUT_MS` (50 ms, roughly 50× the worst-case transfer at 100 kHz), and `OLED_Update()` returns early once the address-window commands fail so a dead panel costs one timeout rather than six.

*A correction to earlier revisions of this report.* Previous revisions recorded the `cpsie / PRIMASK` hypothesis as the leading explanation for this symptom. That hypothesis was **wrong** and is withdrawn: the cause is the critical-section/SysTick gate established above by direct register observation. The three-instruction `msr` workaround in `prvPortStartFirstTask()` is retained because it is harmless on genuine silicon, but it was never the fix. The port as a whole was **adopted** from the reference implementation's design — a SysTick-driven context switch with no PendSV — rather than rejected as earlier revisions stated; the reference's `vPortEnterCritical()` / `vPortExitCritical()` are byte-identical to the ones used here.

*Why the reference implementation is unaffected.* The reference carries the same latent flaw and is only *accidentally* immune: its `main.cpp` creates its mutex and sensor queue and calls `vTaskStartScheduler()` with no `HAL_Delay` and no other HAL-tick dependency in between — there is no `HAL_Delay` anywhere in its `src/`. Nothing that needs the tick runs while the tick is gated, so the permanent gate costs it nothing. This application does need the tick before the scheduler, so the same defect is fatal here.

*One simulator deviation remains, documented and accepted.* Wokwi's Cortex-M3 model does not implement `BASEPRI`, so `portDISABLE_INTERRUPTS()` / `portENABLE_INTERRUPTS()` have no effect there and critical sections are not genuinely protected. This does not affect correctness on real hardware, where `BASEPRI` works as specified, but it does mean a concurrency defect that depends on a critical section would not be caught by a Wokwi run. It is recorded as limitation L-08 in section 7.2, and it is the reason this port gates the tick at its source rather than through the CPU mask.

## 7.4 Alternative Approaches Considered

**Event group vs. task notifications for encoder events:** The design in fact uses *both*. Task notifications carry the wake-up signal from each EXTI ISR to its own task ( `InputTask` and `MotionTask` each block in `ulTaskNotifyTake()` ), because a direct-to-task notification is the cheapest possible ISR-to-task hand-off and is already tied to exactly one consumer. The event group carries the *semantic* event instead: `MOTION_DETECTED_BIT`, `ENCODER_CW_BIT`, `ENCODER_CCW_BIT`, and `ENCODER_BTN_BIT` are set by different producers and observed by `DisplayTask`, which clears all four in one `xEventGroupClearBits()` call. So notifications were chosen for wake-up latency and the event group for multi-producer/multi-bit fan-out.

**DMA for UART output:** The `printf`-based UART output uses blocking HAL transmit ( `HAL_UART_Transmit` ). DMA would free the CPU during transmission but adds complexity. For diagnostic output (not timing-critical), blocking transmit with a mutex is simpler and sufficient.

**Edge-triggered encoder reading vs. periodic polling:** The encoder is read on EXTI edges ( `Encoder_CLK_EXTI_Callback()` in `src/app/hal/encoder.c` ) rather than polled. `InputTask` blocks on `ulTaskNotifyTake(pdTRUE, portMAX_DELAY)` and wakes only when an edge ISR notifies it, so the task consumes no CPU while the encoder is stationary. A 50 ms polling loop would have been simpler, but sampling the quadrature lines that slowly risks missing detents; capturing edges in the ISR and reading a single accumulated delta per wake avoids both the polling cost and the missed-pulse risk.

## 7.5 Diagnostic Instrumentation

Challenge 4 exposed a structural weakness in the firmware that is worth addressing independently of the simulator: **a boot failure was indistinguishable from a hang**. On Cortex-M, the startup file provides every fault handler as a weak alias to a bare infinite loop, so a `HardFault`, a bus fault, or a usage fault all present identically — the board stops producing output and appears dead, with no indication of what went wrong or where. The project supplied no fault handler of its own, so this was the actual behaviour of the firmware as originally submitted.

Three additions address this.

**A fault handler that reports.** `src/drivers/diag.c` installs a `HardFault_Handler` that decodes the stacked exception frame and prints the fault status registers over USART1. It writes to `USART1->DR` directly through a latched register pointer rather than calling `UART_Mutex_Printf()`, because the latter takes a mutex and would deadlock in a fault context — a fault handler cannot assume the scheduler is running or that any lock is free. The handler reads `CFSR`, `HFSR`, `MMFAR`, and `BFAR` to distinguish a precise data-access fault from an imprecise one, and selects the correct stack pointer by testing bit 2 of the link register to determine whether the fault occurred in handler or thread mode. It is declared `naked` so the compiler does not emit a prologue that would corrupt the frame it is trying to read.

**A vector table that is actually relocated.** As described in Challenge 4, the framework's `SystemInit()` is a bare `bx lr` and nothing in the stock boot path programs `VTOR`. The project now calls `Diag_RelocateVectors()` from the top of `main()`, which writes `FLASH_BASE` to `VTOR` before `HAL_Init()` and therefore before the scheduler starts. This is required for the FreeRTOS V11 handler-installation check to read the table it intends to read, and it is what the reference implementation does as well.

The relocation deliberately does **not** take the form of a function named `SystemInit()`. That was the first attempt, and it does not link: the startup file's `b1 SystemInit` resolves against the STM32Cube framework's `system_stm32f1xx.c`, whose `SystemInit` is a plain strong definition rather than `__attribute__((weak))`, so a same-named definition in `src/main.c` is a duplicate symbol at link time (multiple definition of 'SystemInit'). The framework's copy cannot be overridden by name, only shadowed at the object level. Calling the relocation from `main()` instead is functionally equivalent for this design: the only consumer of `VTOR` is the FreeRTOS Cortex-M port, which validates the `SVCALL` and `PendSV` slots in `xPortStartScheduler()` — long after `main()` has started — so the relocation still precedes every read of the vector table. The one property that is lost is that the write no longer happens before the C

runtime copies `.data` and clears `.bss`; that is harmless here because `Diag_RelocateVectors()` writes a constant to a fixed MMIO address and reads no initialised state.

**A boot path that reports its own failure.** `main()` now captures the return code of each `xTaskCreate()` call and prints a fatal message if any of them fails, and prints a second fatal message if `vTaskStartScheduler()` returns — which it can do on heap exhaustion. Previously both conditions were silent. Each of the five task entry points also prints a banner on entry, so the serial log shows exactly how far the scheduler got.

**A visible fault indicator.** `diagram.json` now includes an LED on PC13 with a series resistor. The LED is wired **active-low**, matching the physical Blue Pill: 3.3 V → 330 Ω resistor → LED anode → LED cathode → PC13. The existing `vApplicationStackOverflowHook()` and `vApplicationMallocFailedHook()` toggle that pin forever, so a fault is now visible on the simulated board rather than only in the serial log. In normal operation the LED stays off, because `MX_GPIO_Init()` writes `GPIO_PIN_SET` to PC13 at boot and nothing else drives the pin.

The polarity is worth stating explicitly because it is a real defect that the documentation caught: the LED was first added wired active-high (PC13 → resistor → anode → GND), which is the reverse of the board it is meant to simulate. Since the firmware drives PC13 high at boot, the simulated LED would have been **lit during normal operation** and **dark during a fault** — exactly the inverse of what is wanted, and the opposite of what this report claimed. Wiring it active-low makes the simulated indication agree with the physical board and with the firmware's own polarity.

**A guard against silent fallback.** `tools/verify/check_port_patch.py` runs as a post-build action and verifies four things: that the patched port object is present in the build directory; that `include/portmacro.h` still redirects every macro the kernel core needs; that the linked image contains **no** store to the ICSR at `0xE000ED04` (the PendSV-yield escape described in Challenge 4); and that both critical-section primitives still carry the pre-scheduler guard. It prints a warning rather than failing the build, so that a broken check cannot itself block a compile. All four conditions were confirmed to fail correctly when deliberately broken. Without this, a change to the library configuration could silently revert the port to the unpatched archive member and reintroduce Challenge 4 with no visible sign.

The instrumentation is deliberately dependency-free: `diag.c` includes only `<stdint.h>` and does not use the HAL, FreeRTOS, or the C library, so it remains usable in exactly the situations where those layers are suspect. It adds 1,216 B of `.text` and 36 B of `.bss`, measured with `tools/verify/run_size_analysis.sh`.

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## 8. Conclusion

The BCA182 Room Monitoring System implements a production-quality real-time embedded architecture using FreeRTOS on the STM32F103C8T6. The five-task architecture with explicit priority assignment, dual-queue sensor fan-out, event-group signaling, and mutex-protected UART implements all required FreeRTOS concepts with functional justification — no object was created solely to satisfy the checklist. The design, the hardware-independent logic, and the scheduler's runtime behaviour are all verified: the boot hang that once left the firmware silent has been root-caused, fixed, and replayed successfully, with all five tasks observed running. Physical-hardware validation remains the one item that has not been carried out, for the reason recorded in the last key outcome below.

**Key outcomes:** - All 10 functional requirements (FR-01 through FR-10) are implemented - 33 automated unit tests pass on the native host PC (hardware-independent); these were re-run against the current revision and all pass - The 10 Wokwi functional checks (WF-01 through WF-10) are documented in section 5.2 as development-time observations. Their expected outcomes were verified line by line against the source. A replay against the revision that still contained the scheduler hang reproduced **none** of them; that hang has since been root-caused and fixed (Challenge 4), and a replay of the fixed firmware under Renode shows the scheduler starting and all five tasks running. The WF table therefore records the original per-peripheral observations and is not a current Wokwi result - Static analysis found 0 functional defects across all passes; 21 LOW-severity clang-tidy advisories, 1 compiler warning, and 2 benign memory-mapped-register findings were reviewed and accepted; all re-verified on the current revision - 3 fault experiments reasoned through against the scheduler

rules and checked against the source (section 5.3); their Predicted result paragraphs are derived consequences of the configuration, not measurements - The silent-terminal symptom was root-caused to a critical section taken before `vTaskStartScheduler()`, which permanently closed the port's SysTick gate (`uxCriticalNesting` never returned from its `0xaaaaaaaa` sentinel to 0), and fixed by deferring both critical-section primitives while the scheduler is not started.

**Replayed and verified under Renode:** `uwTick` advances, SysTick is never left gated, and all five tasks are entered (Challenge 4, section 7.3). Two further real defects surfaced in the same replay and are also fixed: a yield through `PendSV` compiled against the stock `portmacro.h`, and a UART mutex reported through before it was created. - The firmware was hardened so that a boot failure is no longer silent: a fault handler that decodes and reports the exception frame, a relocated vector table, captured task-creation return codes, and a visible fault LED (section 7.5)

**Lessons learned:** 1. `vTaskDelayUntil()` is essential for stable periodic sampling; `vTaskDelay()` accumulates drift 2. Shared FreeRTOS queues with multiple consumers require per-consumer copies or depth-1 overwrite queues 3. I2C bulk transfers are orders of magnitude more efficient than per-byte transactions, both on hardware and in simulation 4. Priority assignment must reflect scheduling urgency and consequence of delay — not relative "importance" 5. Every task must contain at least one blocking call to enable cooperative operation of lower-priority tasks 6. A silent failure is worse than a loud one. On Cortex-M the default fault handlers are infinite loops, so a faulting board and a hung board look identical; installing a handler that reports the fault status registers converts an unobservable failure into a diagnosable one 7. A commit message that claims a verified result which was never actually observed is a liability. The 647b343 claim that the OLED rendered in Wokwi was carried in this report for several revisions before the pin-label defect that invalidated it was found (Challenge 2, section 7.3) 8. When a symptom stops exactly at a known boundary — here, at the call to `vTaskStartScheduler()` — that boundary is the most valuable diagnostic clue available, and it is worth eliminating every layer above it before suspecting the layer below. It localised this fault to the port within one pass. 9. When the only available simulator cannot expose the state you need, change simulators rather than reasoning harder. The cause of this hang was invisible in Wokwi — its GDB stub has no hardware watchpoints, so the CPU could not be inspected at the point of failure — and was found within minutes under Renode, which permits registers and memory to be read mid-run. Two further real defects (a `PendSV` yield compiled against the stock header, and a mutex used before it was created) surfaced in the same session. Treat a second, independent execution environment as a debugging instrument, not as a redundancy. 10. An implementation that "works" is not evidence that its design is sound. The reference implementation this project was compared against carries the *same* latent critical-section flaw; it survives only because it never calls `HAL_Delay` before starting its scheduler. Reproducing a working example verbatim would have carried the defect across rather than fixing it. 11. Instrumented failure paths have to be tested by triggering them. The [OLED] `init failed` message added to report display faults was itself faulting — it ran before its mutex existed and tripped a kernel assert. It looked correct for two revisions because the condition that would exercise it (an absent panel) never occurred in the simulator being used. ting the layer below

**Future improvements:** - Add SD card data logging (SPI + FAT filesystem) - Multi-tone buzzer alarms for different alert types - OTA firmware update capability via USART bootloader - Port state machine to use `vTaskPrioritySet()` for dynamic priority adjustment during alarm conditions

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